

DA-Session-10-DPP

Assignment Questions

Question 1:

A company introduces a new customer feedback process. The mean satisfaction score before implementing the process was 72. After implementing the process, a random sample of 30 customers gave a mean satisfaction score of 78, with a standard deviation of 10. At a 5% significance level, test if the new process has improved customer satisfaction.

Answer:

To test whether the new customer feedback process has improved customer satisfaction, we perform a **one-sample t-test**.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The new process has not improved customer satisfaction.

$$H_0: \mu = 72$$

- **Alternative Hypothesis (H_1):** The new process has improved customer satisfaction.

$$H_1: \mu > 72$$

Step 2: Given Data

- Population mean (μ_0) = 72
- Sample mean ($\bar{x}$) = 78
- Sample standard deviation (s) = 10
- Sample size (n) = 30
- Significance level (α) = 0.05

Step 3: Calculate the Test Statistic

$$\begin{aligned} t &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\ t &= \frac{78 - 72}{10/\sqrt{30}} \\ t &= \frac{6}{1.826} \\ t &= 3.29 \end{aligned}$$

Step 4: Critical Value

Degrees of Freedom:

$$df = n - 1 = 29$$

For a one-tailed test at 5% significance level:

$$t_{critical} = 1.699$$

Step 5: Decision

Since

$$t_{calculated} = 3.29 > t_{critical} = 1.699$$

we reject the null hypothesis.

Conclusion

At the **5% significance level**, there is sufficient evidence to conclude that the new customer feedback process has **significantly improved customer satisfaction**.

Final Answer

Since the calculated t-value (3.29) is greater than the critical value (1.699), we reject the null hypothesis. Therefore, the new customer feedback process has significantly improved customer satisfaction at the 5% significance level.

Question 2:

A school claims that their students' average math score is higher than the national average of 75. A random sample of 50 students scored an average of 77 with a standard deviation of 8. Perform a one-tailed test at a 0.01 significance level to verify the school's claim.

Answer:

To verify the school's claim, we perform a **one-tailed hypothesis test**.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The average math score is equal to or less than the national average.

$$H_0: \mu = 75$$

- **Alternative Hypothesis (H_1):** The average math score is higher than the national average.

$$H_1: \mu > 75$$

This is a **right-tailed test**.

Step 2: Given Data

- Population mean (μ_0) = 75
 - Sample mean ($\bar{x}$) = 77
 - Sample standard deviation (s) = 8
 - Sample size (n) = 50
 - Significance level (α) = 0.01
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Step 3: Calculate the Test Statistic

Since the population standard deviation is unknown, use a **one-sample t-test**.

$$\begin{aligned} t &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\ t &= \frac{77 - 75}{8/\sqrt{50}} \\ t &= \frac{2}{1.131} \\ t &\approx 1.77 \end{aligned}$$

Step 4: Determine the Critical Value

Degrees of Freedom:

$$df = n - 1 = 49$$

For a **one-tailed test** at $\alpha = 0.01$ and $df = 49$:

$$t_{critical} \approx 2.405$$

Step 5: Decision

Since

$$t_{\text{calculated}} = 1.77 < t_{\text{critical}} = 2.405$$

we fail to reject the null hypothesis.

Step 6: Conclusion

At the **1% significance level**, there is **insufficient evidence** to support the school's claim that its students' average math score is higher than the national average of 75.

Final Answer

Since the calculated t-value (1.77) is less than the critical value (2.405), we fail to reject the null hypothesis. Therefore, there is not enough evidence at the 0.01 significance level to conclude that the school's average math score is higher than 75.

Question 3:

A retail store wants to check if their new sales strategy increases the average sales from ₹10,000 per day. After implementing the strategy for 15 days, the sales data showed an average of ₹11,200 per day with a standard deviation of ₹1,500. Conduct a one-sample t-test at a 5% significance level.

Answer:

To determine whether the new sales strategy has increased average daily sales, we perform a **one-sample t-test**.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The new strategy has not increased average sales.

$$H_0: \mu = 10000$$

- **Alternative Hypothesis (H_1):** The new strategy has increased average sales.

$$H_1: \mu > 10000$$

This is a **right-tailed test**.

Step 2: Given Data

- Population Mean (μ_0) = ₹10,000
 - Sample Mean ($\bar{x}$) = ₹11,200
 - Sample Standard Deviation (s) = ₹1,500
 - Sample Size (n) = 15
 - Significance Level (α) = 0.05
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Step 3: Calculate the Test Statistic

The t-statistic is calculated using:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substituting the values:

$$\begin{aligned} t &= \frac{11200 - 10000}{1500/\sqrt{15}} \\ &= \frac{1200}{1500/3.873} \\ t &= \frac{1200}{387.30} \\ t &\approx 3.10 \end{aligned}$$

Step 4: Determine the Critical Value

Degrees of Freedom:

$$df = n - 1 = 15 - 1 = 14$$

For a **one-tailed test** at $\alpha = 0.05$ and $df = 14$:

$$t_{critical} \approx 1.761$$

Step 5: Decision Rule

- Reject H_0 if $t > 1.761$
- Otherwise, fail to reject H_0

Since:

$$3.10 > 1.761$$

we **reject the null hypothesis**.

Step 6: Conclusion

At the **5% significance level**, there is sufficient evidence to conclude that the new sales strategy has significantly increased the average daily sales.

Final Answer

Since the calculated t-value (**3.10**) is greater than the critical value (**1.761**), we **reject the null hypothesis**. Therefore, there is sufficient evidence at the 5% significance level to conclude that the new sales strategy has increased the average daily sales.

Question 4:

A factory produces light bulbs with an advertised lifespan of 1200 hours. A sample of 40 bulbs is tested, and the mean lifespan is found to be 1180 hours with a standard deviation of 50 hours. Test if the advertised lifespan is accurate at a 0.05 significance level.

Answer:

To test whether the advertised lifespan of 1200 hours is accurate, we perform a **one-sample t-test**.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The average lifespan of the bulbs is 1200 hours.

$$H_0: \mu = 1200$$

- **Alternative Hypothesis (H_1):** The average lifespan of the bulbs is not 1200 hours.

$$H_1: \mu \neq 1200$$

This is a **two-tailed test** because we are checking whether the advertised lifespan is accurate or not.

Step 2: Given Data

- Population Mean (μ_0) = 1200 hours
 - Sample Mean ($\bar{x}$) = 1180 hours
 - Sample Standard Deviation (s) = 50 hours
 - Sample Size (n) = 40
 - Significance Level (α) = 0.05
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Step 3: Calculate the Test Statistic

The t-statistic is calculated using:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substituting the values:

$$\begin{aligned} t &= \frac{1180 - 1200}{50/\sqrt{40}} \\ &= \frac{-20}{50/6.325} \\ t &= \frac{-20}{7.906} \\ t &\approx -2.53 \end{aligned}$$

Step 4: Determine the Critical Value

Degrees of Freedom:

$$df = n - 1 = 40 - 1 = 39$$

For a **two-tailed test** at $\alpha = 0.05$ and $df = 39$:

$$t_{critical} = \pm 2.023$$

Step 5: Decision Rule

- Reject H_0 if $t < -2.023$ or $t > 2.023$
- Otherwise, fail to reject H_0

Since:

$$-2.53 < -2.023$$

we **reject the null hypothesis**.

Final Answer

Since the calculated t-value (**-2.53**) falls in the rejection region and is less than the critical value (**-2.023**), we **reject the null hypothesis**. Therefore, the advertised lifespan of **1200 hours is not accurate** at the 5% significance level.

Question 5:

A researcher wants to test if the average height of adult males in a city differs from the national average of 5.8 feet. They collected data from 25 individuals and found an average height of 5.7 feet with a standard deviation of 0.3 feet. Perform a two-tailed test at a 1% significance level.

Answer:

To determine whether the average height of adult males in the city differs from the national average, we perform a **two-tailed one-sample t-test**.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The average height is equal to the national average.

$$H_0: \mu = 5.8$$

- **Alternative Hypothesis (H_1):** The average height is different from the national average.

$$H_1: \mu \neq 5.8$$

This is a **two-tailed test**.

Step 2: Given Data

- Population Mean (μ_0) = 5.8 feet
 - Sample Mean ($\bar{x}$) = 5.7 feet
 - Sample Standard Deviation (s) = 0.3 feet
 - Sample Size (n) = 25
 - Significance Level (α) = 0.01
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Step 3: Calculate the Test Statistic

The t-statistic is calculated using:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substituting the values:

$$\begin{aligned} t &= \frac{5.7 - 5.8}{0.3/\sqrt{25}} \\ t &= \frac{-0.1}{0.3/5} \\ t &= \frac{-0.1}{0.06} \\ t &= -1.67 \end{aligned}$$

Step 4: Determine the Critical Value

Degrees of Freedom:

$$df = n - 1 = 25 - 1 = 24$$

For a **two-tailed test** at $\alpha = 0.01$ and $df = 24$:

$$t_{critical} = \pm 2.797$$

Step 5: Decision Rule

- Reject H_0 if $t < -2.797$ or $t > 2.797$
- Otherwise, fail to reject H_0

Since:

$$-2.797 < -1.67 < 2.797$$

we fail to reject the null hypothesis.

Final Answer

Since the calculated t-value (**-1.67**) lies within the acceptance region and does not exceed the critical values (**± 2.797**), we **fail to reject the null hypothesis**. Therefore, there is **insufficient evidence at the 1% significance level** to conclude that the average height of adult males in the city differs from the national average of **5.8 feet**.